

Stepwise Self-Assembly and Dynamic Exchange of Supramolecular Snowflakes

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Dedicated to Professor Makoto Fujita for the Wolf Prize in Chemistry

Abstract: Snowflake, a highly symmetrical hexagram figure, is challenging to be expressed by chemistry/supramolecular chemistry due to the complex structure. Herein, we have constructed super snowflake supramolecules using terpyridine (tpy)-based metal-organic building blocks with <tpy-Ru(II)-tpy> and <tpy-Zn(II)-tpy> connectivities through stepwise strategies in high yield. The structures were characterized by multi-dimensional mass spectrometry and

multi-dimensional NMR spectrometry. In order to address the stability/tolerance of our designed super snowflake structures, ligand exchange behaviors between different supramolecules with various arm length were fully investigated by mass spectrometry. The study revealed that three modes could exist in such binary systems, including full exchange, partial exchange and self-sorting (no exchange) depending on the length difference of ligands.

Keywords: Supramolecule • Terpyridine • Self-assembly • Dynamic Ligand Exchange

1. Introduction

Snowflake, a highly symmetrical hexagram figure has been extensively explored across art, science, mathematics and technology. It is still challenging to construct snowflake (supra)molecules by synthetic approach due to complex structure. Instead of using pure synthetic strategy, we focus our attention on constructing the snowflake structures by utilizing coordination-driven self-assembly approach,^[1] which emerged as a powerful bottom up strategy^[2] in building up various sophisticated two dimensional and three dimensional supramolecular architectures,^[3–16] benefitting from the predictable and highly directional feature of coordination.^[1] Theoretically, the assembled metallo-organic supramolecular structures are predictable and designable by carefully analyzing the bridging angles and the coordination geometry of the ligands and metal centres.^[5b, c] However, due to the flexibility of the chemical bonds and thermal motions of all the composed atoms in structure, the prediction of self-assembled structure becomes dramatically difficult, particularly when elongating the arm length of the bridging ligands to increase the complexity and/or to introduce functional groups into the supramolecular structures.^{[5][17]}

Recently, in order to facilitate the ligand design with precise bond angle and proper arm length, and to prevent unpredicted structures, two shape-persistent snowflake-like supramolecules (**S1** and **S2**)^[18] with all rigid backbone were constructed through stepwise coordination-driven self-assembly strategy using 2,2':6',2''-terpyridine (tpy) ligand, a widely used building block in supramolecular and macromolecular chemistry.^[19–21] They were quite different from the pioneering works of Stang and Newkome on assembling snowflake-like

structures via exo-functionalizing supramolecular hexagons with dendrimers.^[22, 23] To further explore the robustness of our designed super snowflake structure and to enlarge the scope of the possible ligands for future potential functionality, it is extremely intriguing to construct new super snowflake based on longer version of ligands.

Herein, we assembled two larger version of super snowflakes (**S3** and **S4**) with diameter around 10.4 nm and 11.8 nm, respectively, by changing the arm length of the Y components of the ligand, as shown in Scheme 1. Discrete super snowflakes **S3** and **S4** were obtained through stepwise method, and their structures were fully documented through multi-dimensional mass spectrometry and multi-dimensional

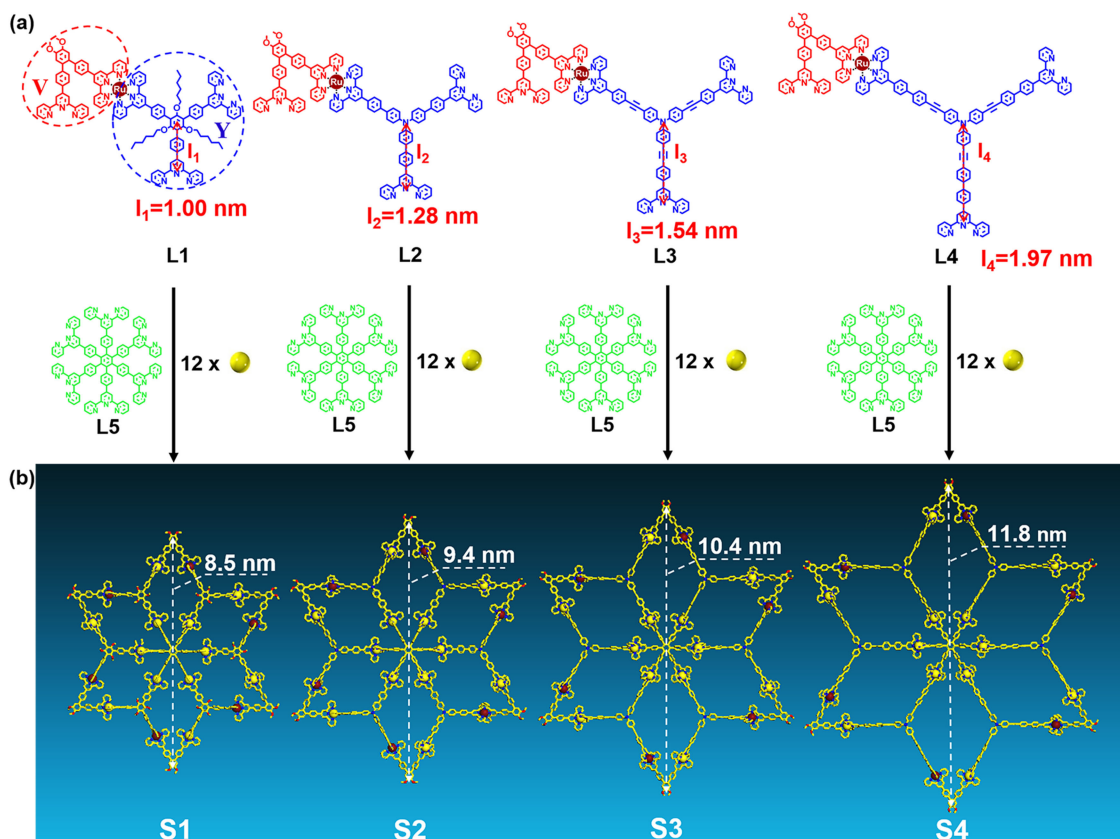
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Scheme 1. Self-assembly of snowflakes **S1**, **S2**, **S3** and **S4**. (a) Structures of ligands **L1**–**L4** with different length of the arms; (b) Energy-minimized structures of snowflakes from molecular modeling. (The alkyl chains are omitted for clarity and I represents the arm length of **Y** component).

NMR spectrometry. Together with previous super snowflakes (**S1** and **S2**),^[18] ligand exchange behaviors between different supramolecules with various arm length (I) were fully investigated. Interestingly, the exchange behaviors varied according to the difference in arm length (ΔI) of the **Y** ligands with three distinct features, including, full exchange, partial exchange and self-sorting (no exchange) status. These phenomena might shed light on studying dynamic behaviors in synthetic/natural supramolecular systems.

2. Results and Discussion

2.1 Synthesis, Self-Assembly and Characterization of **S3** and **S4**

By keeping the rigidity backbone, we introduced one alkyne group on each arm of **Y** component of **L2** to extend the arm length of **L3** (Scheme 1a). **L4** was designed accordingly by adding one additional phenyl group on each arm based on **L3**. As a result, the arm length of each ligand relies on the distance from the tpy group to the core center in its **Y** component. The distance is quantified as I values, which are 1.00 nm, 1.28 nm, 1.54 nm, and 1.97 nm for **L1**–**L4**, respectively (measured

according to theoretical simulation models of the ligands). The size of the self-assembled super snowflakes increases from 8.5 nm (**S1**) and 9.4 nm (**S2**) to 10.4 nm (**S3**) and 11.8 nm (**S4**) based on the simulated models of the structures (Scheme 1b). Compared with **S1**, the diameter of **S3** and **S4** increases around 20 % and 40 %, respectively. We speculated that such increments in size might significantly challenge the robustness of the snowflake structure. Therefore, unpredictable structures might be assembled. Remarkably, the tolerance of the snowflake system is high enough to readily accommodate the variation of the ligand in the outer rim to afford discrete supramolecular structure.

S3 and **S4** were prepared by assembling **L3** and **L4**, respectively, with a hexatopic ligand **L5** and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in exact stoichiometric ratio of 6:1:12 in $\text{CHCl}_3/\text{MeOH}$ mixed solvent (Scheme S3). After incubation at 50 °C for 5 hours, counterion exchange by subsequent treatment with excess amount of NH_4PF_6 gave the complexes **S3** and **S4**. The structures of **S3** and **S4** were then fully characterized by 1D, 2D, and DOSY NMR spectroscopy, electrospray ionization-mass spectrometry (ESI-MS), traveling wave ion mobility-mass spectrometry (TWIM-MS),^[24] and gradient tandem-mass spectrometry gMS².^[25]

NMR characterization was first applied to address the structural information. Figure 1 showed the ^1H NMR spectra of the complexes **S3** and **S4** together with their corresponding ligands. The spectra of **L3**, **L4** and **L5** had sharp and well-splitting signals, among which ligand **L3** and **L4** showed four distinguishable sets of *tpy*-phenyl resonances peaks (A–D) in the spectra. After complexation, the ^1H NMR spectra of **S3** and **S4** became more broad and complicated due to multiple overlapping peaks caused by six sets of *tpy*-phenyl signals

(labeled as A–F). There were no distinguishable shifts in the two sets of peaks for the *tpyH*^{3',5'} protons of <*tpy*-Ru(II)-*tpy*> moieties, but, all the free *tpyH*^{3',5'} protons of **L3**, **L4** and **L5**, i.e., A(E)^{3',5'}, D^{3',5'}, and F^{3',5'}, were shifted to downfield as a result of the lower electron density upon coordination.^[26] In comparison with **L3**, the characteristic upfield shift for the *tpyH*^{6,6''} protons ($\Delta\delta \approx 0.8$ ppm) was observed after the complexation. Similar upfield shifts was detected ($\Delta\delta \approx 0.9$ ppm **S4/L4**) (Figure 1(b)). The peaks assignments were

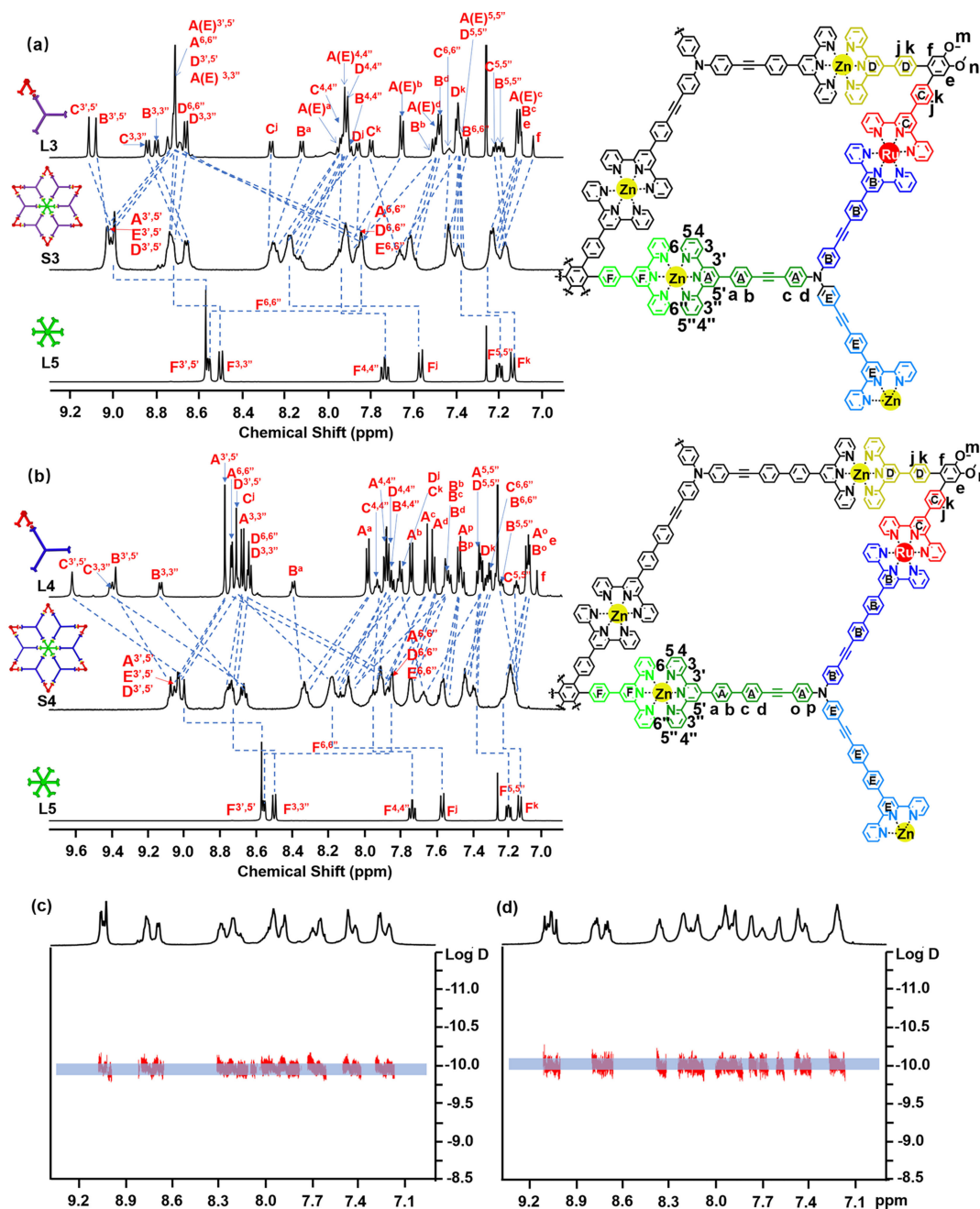


Figure 1. ^1H NMR spectra of ligand and complex. (a) **L3**, **L5** and **S3**; (b) **L4**, **L5** and **S4**; (c) DOSY of **S3**; (d) DOSY of **S4** (CDCl_3 for the ligands and CD_3CN for the supramolecules).

confirmed by the detailed 2D-COSY, 2D-NOESY and 2D-ROESY spectra (Figures S22–S27 and S31–S36). Narrow bands of signals were observed in the 2D diffusion-ordered NMR spectroscopy (DOSY) spectra of **S3**–**S4** (Figure 1(c) and 1(d)), with the diffusion coefficient (D) of 1.11×10^{-10} , $0.98 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ (log D values as -9.95 and -10.01), respectively.

In addition to NMR spectroscopy, ESI-MS coupled with TWIM-MS and gMS² was used to validate the proposed structures. As expected, the complexes were identified as the proposed molecular composition with measured molecular weight at 20484 Da and 21848 Da for **S3** and **S4**, respectively (Figure 2(a) and (d)). Corresponding isotope pattern of each

chargestate agreed well with simulated isotopic distribution (Figures S1 and S2). TWIM-MS was applied to provide additional evidence supporting the structures of **S3** and **S4**. In both cases, linearly and narrowly distributed signal bands on drifting time of various charge states indicated that discrete, shape-persistent supramolecular structures were formed without other conformers or isomers (Figure 2(b) and (e)). Further characterization of **S3** and **S4** on their stabilities was examined by gMS². The 19+ ions of **S3** and **S4** were completely dissociated at 24 V and 21 V in gMS² (Figure 2(c) and (f)), which were converted to center-of-mass collision energy with 0.05 eV and 0.04 eV, respectively. Such difference suggested

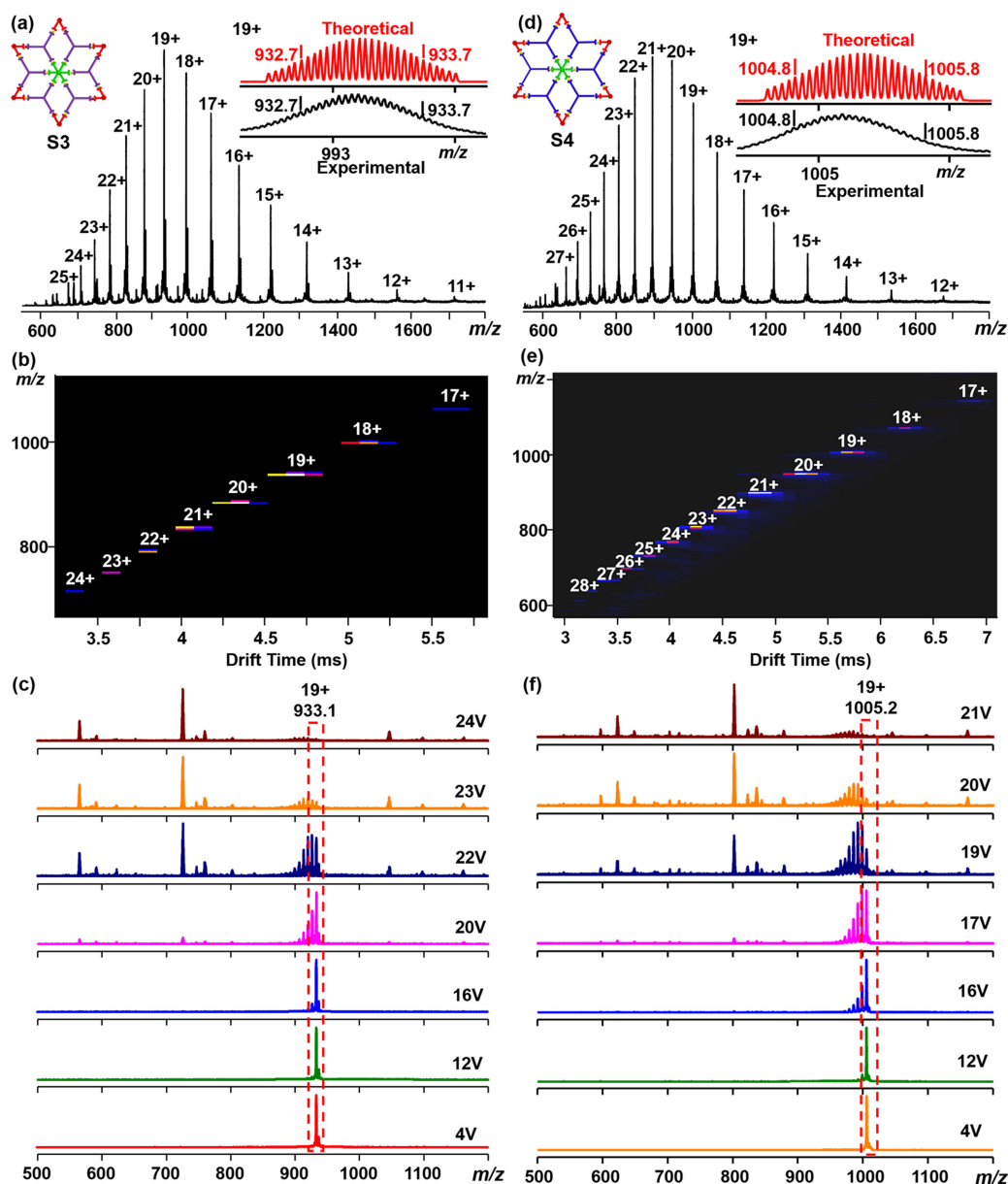


Figure 2. (a and d) ESI-MS and (b and e) TWIM-MS plot (m/z vs. drift time) of **S3** and **S4**. (c) gMS² of **S3** at m/z 933.1 with different collision energy. (f) gMS² of **S4** at m/z 933.1 with different collision energy.

that the stability of **S4** was lower than that of **S3**. Both **S3** and **S4** have same coordination sites, but **S4** has larger size (theoretical collision cross section, CCS) than **S3**. The density of coordination sites (DOCS) calculated by coordination site number/CCS of **S4** is smaller than that of **S3**.^[27] Therefore, the difference of stability of supramolecules is consistent with the difference of DOCS.

2.2 Intermolecular Dynamic Ligand Exchange between Super Snowflakes

The success in preparing larger version of super snowflakes, **S3** and **S4**, sheds some lights on the tolerance/stability of this system, in which the structure is able to accommodate different ligand arm length. In addition, ligand exchange behavior between different snowflakes is another vital aspect of the tolerance in the super structure system. With ligand exchange, hybrid supramolecular structure would accommodate ligands with different arm length by bending some bonds or adjusting some bond angles within one architectures.^[6a,17,28] According to our previous study,^[18] if two candidate ligands are similar enough in structure, for instance, **L1** and **L2**, ligand exchange between their corresponding super snowflakes was readily realized, and formed different hybrid structures in Gaussian statistical distribution. Moreover, we speculate that, by increasing the difference between the ligands, the ligand exchange will become gradually difficult, and might display self-sorting behavior finally rather than ligand exchange. With more ligands in various arm length in hand, we are able to address whether ligand exchange or self-sorting is predominant in the binary mixture of two types of snowflakes.

We carefully compared the ligand exchange behaviors of **S1/S2** (reported),^[18] **S1/S3**, and **S1/S4** systems to address our speculation. The differences of value in Δl in **S1/S2**, **S1/S3**, and **S1/S4** pairs were 0.28 nm, 0.54 nm, and 0.97 nm, respectively.

By increasing Δl from 0.28 nm to 0.54 nm, we mixed preassembled **S1** and **S3** (CH_3CN , 1 mg/mL) solutions with 1:1 ratio at room temperature. The whole experimental process was examined by ESI-MS and TWIM-MS (Figure 3). ESI-MS showed two series of peaks corresponding to **S1** and **S3** at 0 min after mixing (Figure 3(a)). Meanwhile, TWIM-MS showed two clean sets of signals of **S1** and **S3** separately without any superimposition (Figure 3(b)). The slow ligand exchange started around 5 h after the mixing. In addition to the original peaks of **S1** and **S3**, for example **S1**¹⁸⁺ and **S3**¹⁸⁺, five new peaks appeared, consistent with all the statistical hybrid snowflakes, with composition is **Zn₆L₁L₅L₃L₁L₅**, **Zn₆L₁L₃L₅L₃L₅**, **Zn₆L₁L₃L₃L₅L₅**, **Zn₆L₁L₂L₃L₄L₅**, **Zn₆L₁L₁L₃L₅L₅**, respectively. The hybrid peaks increased by extending the mixing time, and finally reach equilibrium (without significant change) at 7 days (Figure 3(e)). In the spectrum of TWIM-MS (Figure 3(d)), all the hybrid signals drifted out with very similar times, indicating that they should have the similar snowflake structure as **S1** and **S3**. Partially exchanging status

was showed when reaching the dynamic equilibrium, because **S1** and **S3** were still observed with high abundance compared to hybrid ones after exchanging in the system. In a sharp contrast, **S1** and **S2** were observed with much less abundance compared to the hybrid ones after exchange in our previous.^[18] As a result, increment in Δl blocked the ligand exchange as expected.

Further increasing Δl to 0.97 nm in **S1/S4** system, two series of peaks corresponding to **S1** and **S4** were observed at 0 min after mixing by ESI-MS and TWIM-MS (Figure 4(a) and 4(b)). However, even after 7 days of mixing, no obvious new peaks were observed by ESI and TWIM-MS (Figure 4(c), 4(d) and 4(e)), suggesting that no exchange occurred when Δl reached a high value. A control study was performed by mixing equal molar amounts of **L1** and **L4** with **L5** and **Zn(II)** at 3:3:1:12 ratio. After heating at 50 °C for 15 hours, the mass spectrometry results showed that two discrete snowflakes coexist without communication (Figure S3).

With Δl as a ruler, we could predict the exchange behavior between two randomly chosen super snowflakes. For instance, in **S3/S4** system, Δl is 0.43 nm, locating between 0.28 nm of **S1/S2** and 0.54 nm of **S1/S3**. Consequently, the abundance of the resultant hybrid super snowflakes compared to pure ones should be less than **S1/S2** case but more than **S1/S3** situation. And ESI-MS results confirmed the prediction, as shown in Figure 5.

2.3 Physical Properties

Considering the **tpy** and triphenylamine have a wide range of applications in optoelectronics,^[29] we studied the photophysical and electrochemical properties of the complexes. In the absorption spectra (Figure 6(a)), **S3** and **S4** showed characteristic absorption band peaked at 500 nm, which corresponds to allowed metal-to-ligand charge transfer (MLCT) transitions.^[30] The band formed around 375 nm was assigned to the intramolecular charge transfer (ICT) transition from triphenylamine moiety to the **tpy**-metal complex moieties.^[31]

Oxidation and reduction properties of two complexes in DMF were determined through cyclic voltammetry (CV). In Figure 6(b), two **tpy**-ligand-centered redox couples observed between −2.0 V and −0.5 V. Between 0.8 V to 1.5 V, **S3** and **S4** showed two irreversible oxidation peaks, corresponding to the oxidization of triphenylamine moiety and **Ru(II)** atoms.

3. Conclusions

In summary, in order to address the stability/tolerance of our designed super snowflake structure and increase the complexity for future functionality, we have constructed a series of super snowflake supramolecules with high symmetry using metal-organic building blocks with <**tpy**-**Ru(II)**-**tpy**> and <**tpy**-**Zn(II)**-**tpy**> motifs through a stepwise strategy. The architectures were fully characterized by 1D and 2D NMR,

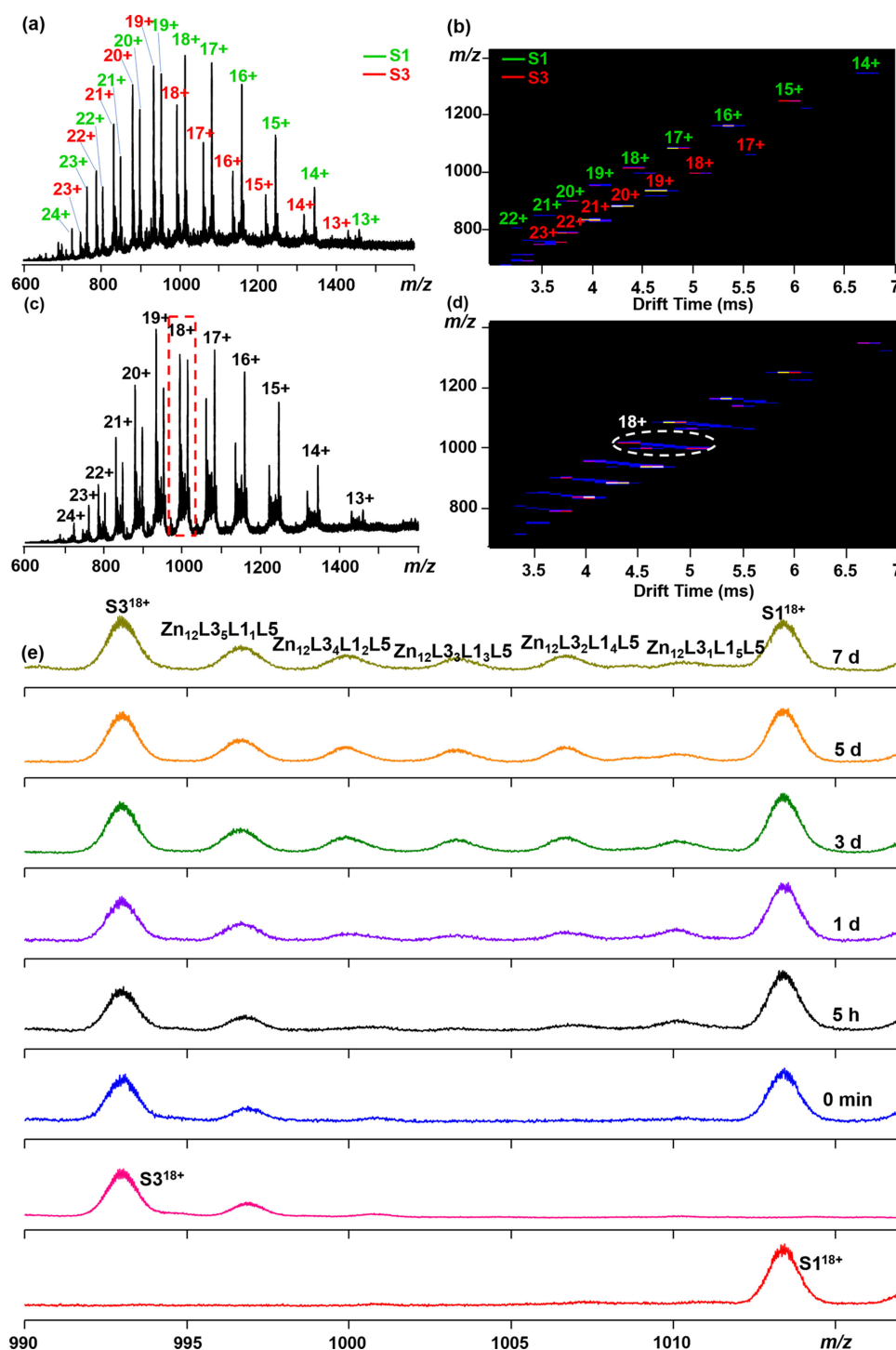


Figure 3. Dynamic exchange of preassembled **S1** and **S3**. (a) ESI-MS and (b) TWIM-MS plot of preassembled **S1** and **S3** at 0 min. (c) ESI-MS and (d) TWIM-MS plot of preassembled **S1** and **S3** at 7 d. (e) Time-dependent spectra of dynamic exchange of **S1** and **S3** at 0 min, 5 h, 1 d, 3 d, 5 d, 7 d for expanded region of 18⁺ signals.

ESI-MS, TWIM-MS, gMS², along with molecular modeling. Furthermore, the dynamic ligand exchange behavior of binary super snowflakes with increasing ΔI values showed full exchange, partial exchanging and no exchange (self-sorting)

status, where ΔI offers us a ruler to predict the exchange behavior of two sets of supramolecules. Such study will guide the rational design and prediction in our further study for more sophisticated with tunable functionality.

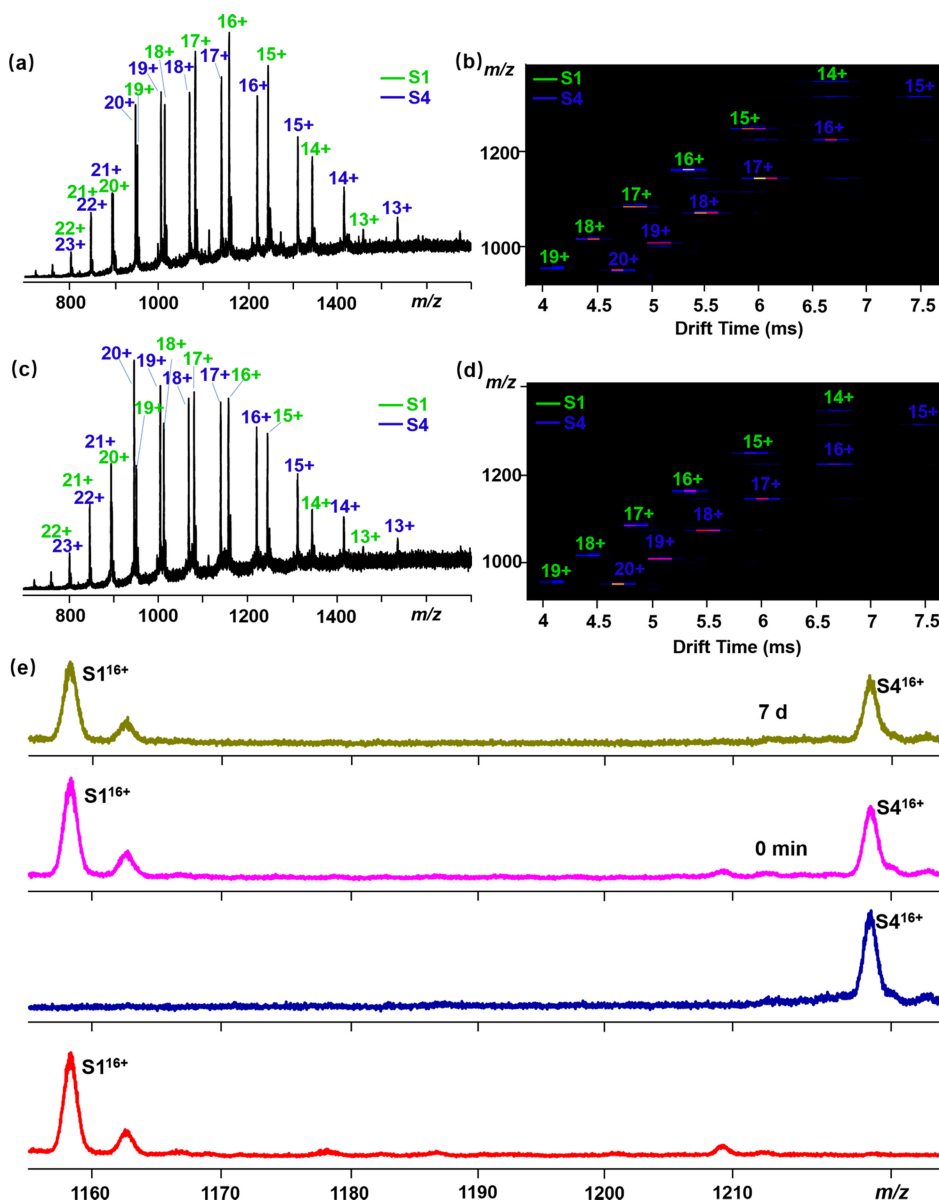


Figure 4. Dynamic exchange of preassembled **S1** and **S4**. (a) ESI-MS and (b) TWIM-MS plot of preassembled **S1** and **S4** at 0 min. (c) ESI-MS and (d) TWIM-MS plot of preassembled **S1** and **S4** at 7 d. (e) Time-dependent spectra of dynamic exchange of **S1** and **S4** at 0 min, 7 d for expanded region of 16+ signals.

4. Experimental Section

All reagents were purchased from Sigma-Aldrich, Matrix Scientific, Alfa Aesar and used without further purification. Column chromatography was conducted using basic Al_2O_3 (Brockman I, activity, 58 Å) or SiO_2 (VWR, 40–60 µm, 60 Å) and the separated products were visualized by UV light or detected by ESI-TOF. NMR spectra data were recorded on a 500-MHz and 600 MHz Bruker Avance NMR spectrometer in CDCl_3 , CD_3CN , CD_3OD . ESI-MS and TWIM-MS were recorded with a Waters Synapt G2 tandem mass spectrometer, using solutions of 0.01 mg sample in 1 mL of $\text{CHCl}_3/\text{CH}_3\text{OH}$

(1:3, v/v) for ligands or 0.5 mg sample in 1 mL of MeCN/MeOH (3:1, v/v) for complexes. MALDI-TOF was carried out on Bruker AutoFlex TOF/TOF mass spectrometer.

Synthesis of Ligand L3. Compound **10** (48.40 mg, 0.025 mmol), compound **1** (35.5 mg, 0.1 mmol), $\text{Pd}(\text{PPh}_3)_4$ (14.44 mg, 0.0125 mmol), potassium carbonate (138 mg, 0.1 mmol) were added into a 100 mL Schlenk flask. 15 mL THF, 3.75 mL MeCN , 7.5 mL H_2O were added under N_2 . The mixture was stirred at 75 °C for 24 h. After cooling to room temperature, the mixture was extracted with CHCl_3 . The combined organic layer was washed with brine, dried over anhydrous Na_2SO_4 , and then concentrated in vacuo, the crude

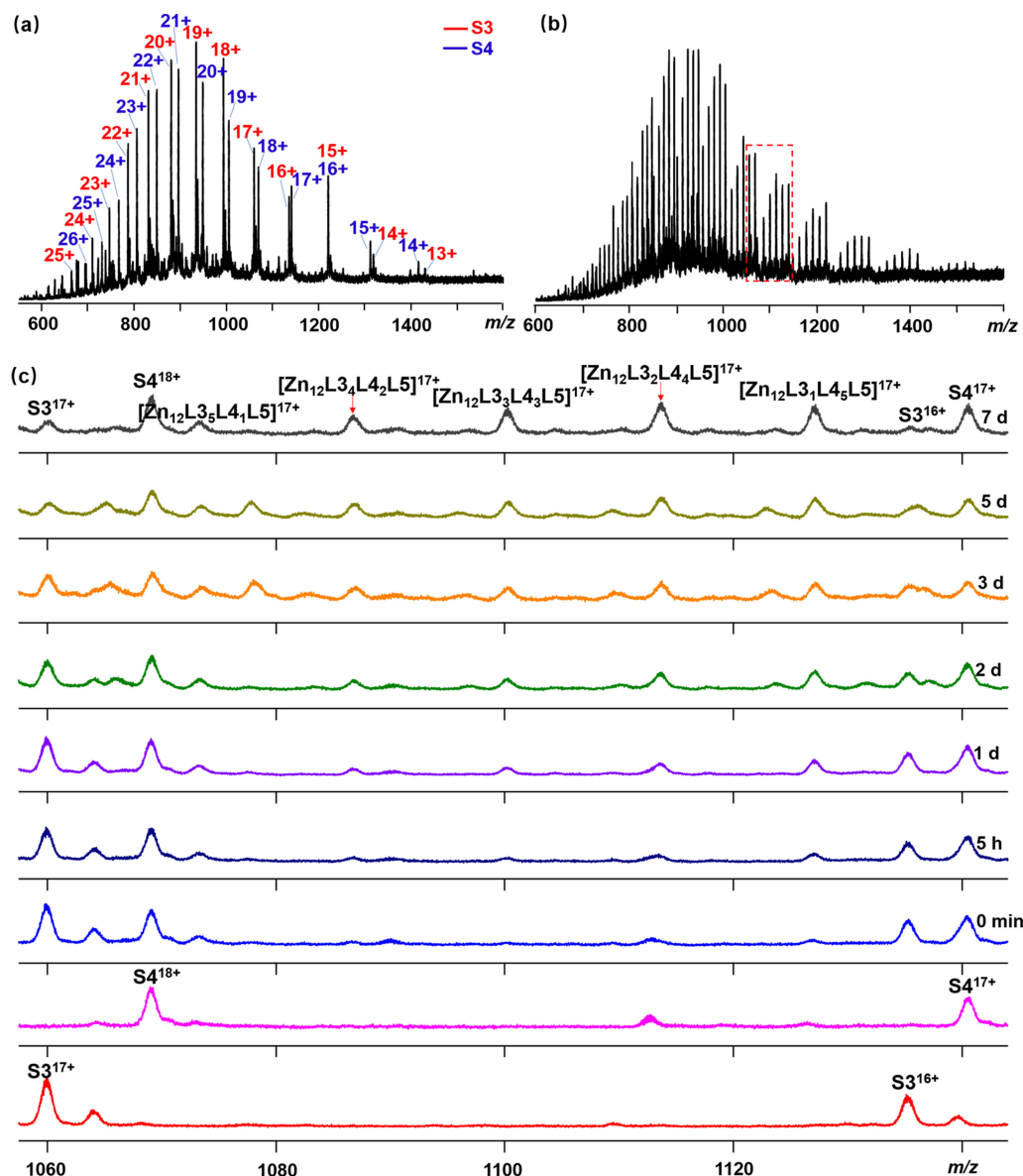


Figure 5. Dynamic exchange of preassembled **S3** and **S4**. (a and b) ESI-MS of preassembled **S3** and **S4** at 0 min and 7 d. (c) Time-dependent spectra of dynamic exchange of **S3** and **S4** at 0 min, 5 h, 1 d, 2 d, 3 d, 5 d, 7 d for expanded region of 17+ signals.

product was purified by column chromatography on silica gel with chloroform: methanol (100:1.7) as eluent to afford the product as a red solid (39 mg, 72.1%). ^1H NMR (600 MHz, CDCl_3 , 300 K, ppm): δ 9.12 (s, 2H, $\text{tpy}^{\text{C}}\text{-H}^{3,5'}$), 9.08 (s, 2H, $\text{tpy}^{\text{B}}\text{-H}^{3,5'}$), 8.84 (d, $J=8.1$ Hz, 2H, $\text{tpy}^{\text{C}}\text{-H}^{3,3''}$), 8.80 (d, $J=8.1$ Hz, 2H, $\text{tpy}^{\text{B}}\text{-H}^{3,3''}$), 8.76–8.68 (m, 14H, $\text{tpy}^{\text{A}}\text{-H}^{3,5'}$, $\text{tpy}^{\text{A}}\text{-H}^{6,6''}$, $\text{tpy}^{\text{D}}\text{-H}^{3,5'}$ and $\text{tpy}^{\text{A}}\text{-H}^{3,3''}$), 8.66 (d, $J=7.9$ Hz, 4H, $\text{tpy}^{\text{D}}\text{-H}^{6,6''}$ and $\text{tpy}^{\text{D}}\text{-H}^{3,3''}$), 8.26 (d, $J=8.3$ Hz, 2H, $\text{Ph}^{\text{C}}\text{-H}^{\text{f}}$), 8.12 (d, $J=8.2$ Hz, 2H, $\text{Ph}^{\text{B}}\text{-H}^{\text{a}}$), 8.02–7.88 (m, 14H, $\text{Ph}^{\text{A}}\text{-H}^{\text{a}}$, $\text{tpy}^{\text{C}}\text{-H}^{4,4''}$, $\text{tpy}^{\text{D}}\text{-H}^{4,4''}$, $\text{tpy}^{\text{A}}\text{-H}^{4,4''}$ and $\text{tpy}^{\text{B}}\text{-H}^{4,4''}$), 7.86 (d, $J=8.0$ Hz, 2H, $\text{Ph}^{\text{D}}\text{-H}^{\text{f}}$), 7.80 (s, 2H, $\text{Ph}^{\text{C}}\text{-H}^{\text{g}}$), 7.65 (d, $J=8.4$ Hz, 4H, $\text{Ph}^{\text{A}}\text{-H}^{\text{b}}$), 7.52–7.46 (m, 8H, $\text{Ph}^{\text{B}}\text{-H}^{\text{b}}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{d}}$ and $\text{Ph}^{\text{B}}\text{-H}^{\text{d}}$), 7.46–7.37 (m, 10H, $\text{tpy}^{\text{C}}\text{-H}^{6,6''}$, $\text{Ph}^{\text{D}}\text{-H}^{\text{g}}$, $\text{tpy}^{\text{A}}\text{-H}^{5,5'}$ and $\text{tpy}^{\text{D}}\text{-H}^{5,5'}$), 7.35 (s, 2H, $\text{tpy}^{\text{B}}\text{-H}^{6,6''}$), 7.23–7.17 (m, 4H, $\text{tpy}^{\text{C}}\text{-H}^{5,5'}$ and $\text{tpy}^{\text{B}}\text{-H}^{5,5'}$), 7.10 (dd, $J=7.0$, 5.5 Hz, 7H, $\text{Ph}^{\text{A}}\text{-H}^{\text{c}}$, $\text{Ph}^{\text{B}}\text{-H}^{\text{c}}$ and $\text{Ph}^{\text{H}}\text{-H}^{\text{c}}$), 7.04 (s, 1H, $\text{Ph}^{\text{H}}\text{-H}^{\text{f}}$), 4.02 (s, 3H, $-\text{OCH}_3\text{-H}^{\text{b}}$), 4.00 (s, 3H, $-\text{OCH}_3\text{-H}^{\text{m}}$). ^{13}C NMR (125 MHz, CD_3CN , 300 K, ppm): δ 158.27, 158.16, 155.80, 155.52, 155.43, 152.28, 152.00, 150.00, 149.16, 149.07, 148.91, 147.38, 147.07, 144.31, 143.00, 138.81, 138.69, 138.18, 137.87, 135.83, 134.37, 133.30, 133.25, 132.96, 132.85, 132.48, 132.30, 131.66, 131.14, 128.32, 128.20, 127.83, 127.51, 127.35, 126.26, 125.58, 125.37, 124.92, 124.72, 124.60, 124.49, 124.26, 122.26, 121.93, 121.77, 119.28, 118.25, 117.71, 114.05, 114.00, 92.52, 91.23, 89.36, 56.54, 56.49. MALDI-TOF MS (m/z): Calcd. for $[\text{C}_{137}\text{H}_{90}\text{N}_{16}\text{O}_2\text{RuCl}]^+$: 2127.6. Found:

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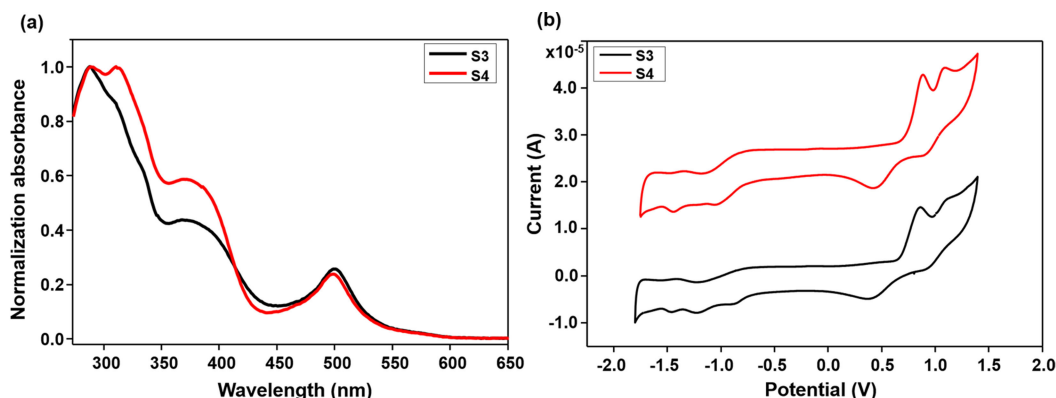


Figure 6. (a) UV-Vis (10^{-6} M in DMF, RT) of complexes of **S3** and **S4**. (b) Cyclic voltammograms (CV) of solutions of **S3** and **S4** (performed in 0.1 M n-Bu₄NBF₄ in DMF at 298 K).

2127.6; Calcd. for [C₁₃₇H₁₁₇ N₁₅O₅Ru + e⁻]⁺: 2092.6. Found: 2092.6.

Synthesis of Ligand L4. Compound **13** (108.21 mg, 0.05 mmol), compound **1** (70.6 mg, 0.2 mmol), Pd(PPh₃)₄ (28.8 mg, 0.025 mmol), potassium carbonate (276 mg, 0.2 mmol) were added into a 100 mL Schlenk flask. 30 mL THF, 7 mL MeCN, 15 mL H₂O were added under N₂. The mixture was stirred at 75 °C for 24 h. After cooling to room temperature, the mixture was extracted with CHCl₃. The combined organic layer was washed with brine, dried over anhydrous Na₂SO₄, and then concentrated in vacuo, the crude product was purified by column chromatography on silica gel with chloroform: methanol (100:1) as eluent to afford the product as a red solid (75 mg, 69.4%). ¹H NMR (600 MHz, CDCl₃, 300 K, ppm): δ 9.62 (s, 2H, tpy^C-H^{3',5'}), 9.43–9.37 (m, 4H, tpy^C-H^{3,3''} and tpy^B-H^{3',5'}), 9.13 (d, *J* = 8.2 Hz, 2H, tpy^B-H^{3,3''}), 8.77 (s, 4H, tpy^A-H^{3',5'}), 8.74 (d, *J* = 3.9 Hz, 4H, tpy^A-H^{6,6''}), 8.71 (d, *J* = 5.7 Hz, 4H, tpy^D-H^{3',5'} and Ph^C-Hⁱ), 8.68 (d, *J* = 7.9 Hz, 4H, tpy^A-H^{3,3''}), 8.64 (t, *J* = 6.1 Hz, 4H, tpy^A-H^{6,6''} and tpy^D-H^{3,3''}), 8.39 (d, *J* = 8.1 Hz, 2H, Ph^B-H^a), 7.98 (d, *J* = 8.3 Hz, 4H, Ph^A-H^a), 7.93 (t, *J* = 7.4 Hz, 2H, tpy^C-H^{4,4''}), 7.89–7.84 (m, 8H, tpy^B-H^{4,4''}, tpy^A-H^{4,4''} and tpy^D-H^{4,4''}), 7.80 (t, *J* = 7.3 Hz, 4H, Ph^A-H^b), 7.74 (d, *J* = 8.3 Hz, 4H, Ph^D-Hⁱ and Ph^C-H^k), 7.66 (d, *J* = 8.4 Hz, 4H, Ph^A-H^c), 7.62 (d, *J* = 8.4 Hz, 4H, Ph^A-H^d), 7.54 (dd, *J* = 18.7, 8.1 Hz, 6H, Ph^B-H^b, Ph^B-H^c and Ph^B-H^d), 7.46 (dd, *J* = 11.9, 8.7 Hz, 6H, Ph^A-H^b and Ph^B-H^b), 7.38–7.29 (m, 12H, tpy^A-H^{5,5''}, tpy^D-H^{5,5''}, Ph^D-H^k, tpy^B-H^{6,6''} and tpy^C-H^{6,6''}), 7.24 (d, *J* = 6.5 Hz, 2H, tpy^B-H^{5,5''}), 7.17–7.14 (m, 2H, tpy^C-H^{5,5''}), 7.11–7.07 (m, 7H, Ph^H^c, Ph^B-H^c and Ph^A-H^c), 7.04 (s, 1H, Ph^H^f), 4.00 (d, *J* = 5.6 Hz, 6H, –OCH₃-H^m and –OCH₃-H^m). ¹³C NMR (125 MHz, CDCl₃, 300 K, ppm): δ 158.39, 158.11, 156.36, 156.28, 156.13, 156.07, 155.26, 155.20, 151.80, 149.88, 149.70, 149.28, 149.18, 148.89, 148.83, 148.68, 146.79, 144.15, 142.59, 142.05, 141.03, 140.07, 139.36, 138.71, 138.50, 137.71, 137.17, 137.03, 136.81, 136.34, 135.05, 133.75, 133.03, 132.63, 132.18, 132.08, 131.56, 130.84, 129.23, 128.06, 127.93, 127.56, 127.12, 127.02, 126.52, 125.89, 124.28, 124.12, 124.00, 123.20, 122.81, 122.28, 121.88, 121.60, 121.53, 118.78,

118.66, 118.22, 117.82, 113.87, 90.39, 89.48, 56.50, 56.35. ESI-TOF (*m/z*): Calcd. for [C₁₅₅H₁₀₂N₁₆O₂BrRuCl₂–2Cl]²⁺: 1160.85. Found: 1160.82; Calcd. for [C₁₅₅H₁₀₂N₁₆O₂BrRuCl₂–2Cl + H]³⁺: 774.23. Found: 774.20.

Synthesis of S3 [Zn₁₂L₃L₅]. To a solution of ligand **L3** (5.0 mg, 2.1 μmol) and **L5** (0.672 mg, 0.35 μmol) in CHCl₃ (2 mL), a solution of Zn(NO₃)₂·6H₂O (1.25 mg, 4.2 μmol) in MeOH (6 mL) was added. The mixture was stirred at 50 °C for 5 h and then cooled to room temperature. Upon addition of NH₄PF₆ (50 mg), a precipitate was formed and washed with water, dried in vacuum to give the product as a red solid (6.9 mg, 96.4%). ESI-MS (*m/z*): 1562.1 [M–12PF₆]¹²⁺ (calcd *m/z*: 1562.0), 1430.7 [M–13PF₆]¹³⁺ (calcd *m/z*: 1430.7), 1318.2 [M–14PF₆]¹⁴⁺ (calcd *m/z*: 1318.2), 1220.5 [M–15PF₆]¹⁵⁺ (calcd *m/z*: 1220.5), 1135.2 [M–16PF₆]¹⁶⁺ (calcd *m/z*: 1135.2), 1059.9 [M–17PF₆]¹⁷⁺ (calcd *m/z*: 1059.9), 993.1 [M–18PF₆]¹⁸⁺ (calcd *m/z*: 993.1), 933.1 [M–19PF₆]¹⁹⁺ (calcd *m/z*: 933.1), 879.1 [M–20PF₆]²⁰⁺ (calcd *m/z*: 879.1), 830.3 [M–21PF₆]²¹⁺ (calcd *m/z*: 830.3), 786.0 [M–22PF₆]²²⁺ (calcd *m/z*: 786.0), 745.5 [M–23PF₆]²³⁺ (calcd *m/z*: 745.5), 708.5 [M–24PF₆]²⁴⁺ (calcd *m/z*: 708.5). ¹H NMR (600 MHz, CD₃CN, 300 K, ppm): δ 9.07–8.97 (m, 12H, tpy^B-H^{3',5'}, tpy^E-H^{3',5'}, tpy^A-H^{3',5'}, tpy^C-H^{3',5'}, tpy^D-H^{3',5'} and tpy^F-H^{3',5'}), 8.74 (br, 8H, tpy^E-H^{3,3''}, tpy^A-H^{3,3''}, tpy^D-H^{3,3''} and tpy^F-H^{3,3''}), 8.66 (d, *J* = 7.0 Hz, 4H, tpy^B-H^{3,3''} and tpy^C-H^{3,3''}), 8.30–8.11 (m, 18H, Ph^C-Hⁱ, Ph^B-H^a, Ph^A-H^a, Ph^E-H^a, Ph^D-Hⁱ, Ph^F-Hⁱ, tpy^A-H^{4,4''}, tpy^D-H^{4,4''} and tpy^E-H^{4,4''}), 7.93 (br, 12H, tpy^B-H^{4,4''}, tpy^C-H^{4,4''}, tpy^F-H^{4,4''}, Ph^A-H^b, Ph^E-H^b and Ph^B-H^b), 7.84 (br, 8H, tpy^A-H^{6,6''}, tpy^E-H^{6,6''}, tpy^D-H^{6,6''} and tpy^F-H^{6,6''}), 7.64 (br, 10H, Ph^C-H^k, Ph^D-H^k, Ph^B-H^d, Ph^A-H^d and Ph^E-H^d), 7.41 (br, 12H, tpy^C-H^{6,6''}, tpy^B-H^{6,6''}, tpy^A-H^{5,5''}, tpy^E-H^{5,5''}, tpy^D-H^{5,5''} and tpy^F-H^{5,5''}), 7.27–7.16 (m, 14H, Ph^F-H^k, Ph^B-H^c, Ph^A-H^c, Ph^E-H^c, Ph^H^c, tpy^C-H^{5,5''} and tpy^B-H^{5,5''}), 4.02 (s, 6H). ¹³C NMR (125 MHz, CD₃CN, 300 K, ppm): δ 158.16, 157.06, 155.41, 152.41, 149.93, 147.88, 147.17, 141.21, 138.02, 135.74, 133.03, 132.43, 131.37, 128.36, 128.02, 127.51, 124.54, 123.29, 121.46, 121.28, 55.80.

Synthesis of S4 [Zn₁₂L₄L₅]. To a solution of ligand **L4** (5.0 mg, 1.9 μmol) and **L5** (0.613 mg, 0.32 μmol) in CHCl₃

(2 mL), a solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1.14 mg, 3.8 μmol) in MeOH (6 mL) was added. The mixture was stirred at 50 °C for 5 h and then cooled to room temperature. Upon addition of NH_4PF_6 (50 mg), a precipitate was formed and washed with water, dried in vacuum to give the product as a red solid (6.6 mg, 94.7%). ESI-MS (m/z): 1675.7 $[\text{M}-12\text{PF}_6]^{12+}$ (calcd m/z : 1675.7), 1536.1 $[\text{M}-13\text{PF}_6]^{13+}$ (calcd m/z : 1536.1), 1416.1 $[\text{M}-14\text{PF}_6]^{14+}$ (calcd m/z : 1416.1), 1312.0 $[\text{M}-15\text{PF}_6]^{15+}$ (calcd m/z : 1312.0), 1220.8 $[\text{M}-16\text{PF}_6]^{16+}$ (calcd m/z : 1220.8), 1140.5 $[\text{M}-17\text{PF}_6]^{17+}$ (calcd m/z : 1140.5), 1069.1 $[\text{M}-18\text{PF}_6]^{18+}$ (calcd m/z : 1069.1), 1005.2 $[\text{M}-19\text{PF}_6]^{19+}$ (calcd m/z : 1005.2), 947.7 $[\text{M}-20\text{PF}_6]^{20+}$ (calcd m/z : 947.7), 895.6 $[\text{M}-21\text{PF}_6]^{21+}$ (calcd m/z : 895.6), 848.3 $[\text{M}-22\text{PF}_6]^{22+}$ (calcd m/z : 848.3), 805.1 $[\text{M}-23\text{PF}_6]^{23+}$ (calcd m/z : 805.1), 765.5 $[\text{M}-24\text{PF}_6]^{24+}$ (calcd m/z : 765.5), 729.1 $[\text{M}-25\text{PF}_6]^{25+}$ (calcd m/z : 729.1), 695.5 $[\text{M}-26\text{PF}_6]^{26+}$ (calcd m/z : 695.5), 664.3 $[\text{M}-27\text{PF}_6]^{27+}$ (calcd m/z : 664.3). ^1H NMR (600 MHz, CD_3CN , 300 K, ppm): δ 9.11–8.97 (m, 12H, $\text{tpy}^{\text{B}}\text{-H}^{3,5'}$, $\text{tpy}^{\text{E}}\text{-H}^{3,5'}$, $\text{tpy}^{\text{A}}\text{-H}^{3,5'}$, $\text{tpy}^{\text{C}}\text{-H}^{3,5'}$, $\text{tpy}^{\text{D}}\text{-H}^{3,5'}$ and $\text{tpy}^{\text{F}}\text{-H}^{3,5'}$), 8.77–8.65 (m, 12H, $\text{tpy}^{\text{E}}\text{-H}^{3,3'}$, $\text{tpy}^{\text{A}}\text{-H}^{3,3'}$, $\text{tpy}^{\text{D}}\text{-H}^{3,3'}$, $\text{tpy}^{\text{F}}\text{-H}^{3,3'}$, $\text{tpy}^{\text{B}}\text{-H}^{3,3'}$ and $\text{tpy}^{\text{C}}\text{-H}^{3,3'}$), 8.32 (br, $\text{Ph}^{\text{B}}\text{-H}^{\text{a}}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{a}}$ and $\text{Ph}^{\text{E}}\text{-H}^{\text{a}}$), 8.22–8.06 (m, 18H, $\text{Ph}^{\text{C}}\text{-H}^{\text{j}}$, $\text{Ph}^{\text{D}}\text{-H}^{\text{j}}$, $\text{Ph}^{\text{F}}\text{-H}^{\text{j}}$, $\text{tpy}^{\text{A}}\text{-H}^{4,4'}$, $\text{tpy}^{\text{D}}\text{-H}^{4,4'}$, $\text{tpy}^{\text{E}}\text{-H}^{4,4'}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{b}}$, $\text{Ph}^{\text{E}}\text{-H}^{\text{b}}$ and $\text{Ph}^{\text{B}}\text{-H}^{\text{b}}$), 7.98–7.82 (m, 20H, $\text{tpy}^{\text{B}}\text{-H}^{4,4'}$, $\text{py}^{\text{C}}\text{-H}^{4,4'}$, $\text{tpy}^{\text{F}}\text{-H}^{4,4'}$, $\text{Ph}^{\text{B}}\text{-H}^{\text{c}}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{c}}$, $\text{Ph}^{\text{E}}\text{-H}^{\text{c}}$, $\text{tpy}^{\text{A}}\text{-H}^{6,6'}$, $\text{tpy}^{\text{E}}\text{-H}^{6,6'}$, $\text{tpy}^{\text{D}}\text{-H}^{6,6'}$ and $\text{tpy}^{\text{F}}\text{-H}^{6,6'}$), 7.74 (br, 6H, $\text{Ph}^{\text{B}}\text{-H}^{\text{d}}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{d}}$, and $\text{Ph}^{\text{E}}\text{-H}^{\text{d}}$), 7.67 (br, 4H, $\text{Ph}^{\text{C}}\text{-H}^{\text{k}}$ and $\text{Ph}^{\text{D}}\text{-H}^{\text{k}}$), 7.57 (br, 6H, $\text{Ph}^{\text{B}}\text{-H}^{\text{p}}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{p}}$, and $\text{Ph}^{\text{E}}\text{-H}^{\text{p}}$), 7.48–7.36 (m, 12H, $\text{tpy}^{\text{C}}\text{-H}^{6,6'}$, $\text{tpy}^{\text{B}}\text{-H}^{6,6'}$, $\text{tpy}^{\text{A}}\text{-H}^{5,5'}$, $\text{tpy}^{\text{E}}\text{-H}^{5,5'}$, $\text{tpy}^{\text{D}}\text{-H}^{5,5'}$ and $\text{tpy}^{\text{F}}\text{-H}^{5,5'}$), 7.26–7.13 (m, 14H, $\text{Ph}^{\text{F}}\text{-H}^{\text{k}}$, $\text{Ph}^{\text{H}}\text{-H}^{\text{e}}$, $\text{Ph}^{\text{H}}\text{-H}^{\text{f}}$, $\text{Ph}^{\text{B}}\text{-H}^{\text{o}}$, $\text{Ph}^{\text{A}}\text{-H}^{\text{o}}$, $\text{Ph}^{\text{E}}\text{-H}^{\text{o}}$, $\text{tpy}^{\text{C}}\text{-H}^{5,5'}$ and $\text{tpy}^{\text{B}}\text{-H}^{5,5'}$), 4.02 (s, 6H). ^{13}C NMR (125 MHz, CD_3CN , 300 K, ppm): δ 158.22, 155.50, 149.87, 149.32, 147.92, 147.58, 141.26, 138.02, 134.60, 134.34, 134.27, 132.86, 132.08, 131.40, 130.30, 130.06, 128.76, 128.41, 128.00, 127.51, 127.37, 126.71, 126.63, 124.68, 124.45, 123.50, 122.99, 121.41, 102.84, 102.42, 56.40.

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